DBMS LAB 5 — ADVANCED SQL QUERIES

39 Questions with Oracle SQL Query Answers

Topics: Aggregate Functions, GROUP BY, HAVING, JOIN, Nested Queries, String Functions, Date Functions, Numeric Functions, Views, Indexes, Correlated Subqueries, NOT EXISTS and Set Operations.

Schema used from the lab manual: DEPT, EMP, PROJ, TASK and DEPENDENT. The manual defines PROJ with PRJ_CREDIT rather than a separate BUDGET column, so Q16 and Q36 treat PRJ_CREDIT as the project budget. The EMP schema contains one PRJ_ID column; Q25 therefore counts distinct PRJ_ID values per employee based on the supplied schema.

1. List the total number of employees.

```
SELECT COUNT(*) AS total_employees
FROM EMP;
```

2. List the minimum, maximum and average salaries and total salary.

```
SELECT MIN(SAL) AS min_sal,
       MAX(SAL) AS max_sal,
       AVG(SAL) AS avg_sal,
       SUM(SAL) AS total_salary
FROM EMP;
```

3. List the Project names undertaken by Marketing Department.

```
SELECT P.PRJ_NAME
FROM PROJ P
JOIN DEPT D ON P.DNO = D.DNO
WHERE UPPER(D.DNAME) = 'MARKETING';
```

4. Display employee names in capital, lower, 1st character capital, number of characters and 3 characters from 2nd position.

```
SELECT ENAME,
       UPPER(ENAME) AS upper_name,
       LOWER(ENAME) AS lower_name,
       INITCAP(ENAME) AS initcap_name,
       LENGTH(ENAME) AS name_length,
       SUBSTR(ENAME, 2, 3) AS three_from_second
FROM EMP;
```

5. List the names of employees who are working under manager 'Raghu'.

```
SELECT ENAME
FROM EMP
WHERE MGRID = (
    SELECT EMPNO
    FROM EMP
    WHERE UPPER(ENAME) = 'RAGHU'
);
```

6. Display department name, Max salary and Min salary in each department.

```
SELECT D.DNAME,
       MAX(E.SAL) AS max_salary,
       MIN(E.SAL) AS min_salary
FROM DEPT D
JOIN EMP E ON D.DNO = E.DEPTNO
GROUP BY D.DNAME;
```

7. Display number of employees working in each department and their department name.

```
SELECT D.DNAME,
       COUNT(E.EMPNO) AS employee_count
FROM DEPT D
LEFT JOIN EMP E ON D.DNO = E.DEPTNO
GROUP BY D.DNAME;
```

8. List the employees whose experience is more than 5 years.

```
SELECT *
FROM EMP
WHERE MONTHS_BETWEEN(SYSDATE, DATE_JOIN) > 60;
```

9. List employees born in December 1984.

```
SELECT *
FROM EMP
WHERE TO_CHAR(DATE_BIRTH, 'MM') = '12'
  AND TO_CHAR(DATE_BIRTH, 'YYYY') = '1984';
```

10. List the number of employees per job title.

```
SELECT JOB,
       COUNT(*) AS employee_count
FROM EMP
GROUP BY JOB;
```

11. List the departments with more than three employees.

```
SELECT D.DNAME,
       COUNT(E.EMPNO) AS employee_count
FROM DEPT D
JOIN EMP E ON D.DNO = E.DEPTNO
GROUP BY D.DNAME
HAVING COUNT(E.EMPNO) > 3;
```

12. List the employees working under the project 'Payroll Automation'.

```
SELECT E.ENAME
FROM EMP E
JOIN PROJ P
  ON E.DEPTNO = P.DNO
 AND E.PRJ_ID = P.PRJ_NO
WHERE UPPER(P.PRJ_NAME) = 'PAYROLL AUTOMATION';
```

13. Illustrate CEIL, FLOOR, TRUNCATE and ROUND on the value 2.83.

```
SELECT CEIL(2.83)  AS ceiling_value,
       FLOOR(2.83) AS floor_value,
       TRUNC(2.83) AS truncate_value,
       ROUND(2.83) AS round_value
FROM DUAL;
```

14. List the projects which have duration of more than 1 year.

```
SELECT PRJ_NO, PRJ_NAME, STRT_DATE, END_DATE
FROM PROJ
WHERE MONTHS_BETWEEN(END_DATE, STRT_DATE) > 12;
```

15. List all employees along with the tasks they are associated with through their department's projects.

```
SELECT E.ENAME,
       P.PRJ_NAME,
       T.TASK_NAME,
       T.STATUS
FROM EMP E
JOIN PROJ P
  ON E.DEPTNO = P.DNO
 AND E.PRJ_ID = P.PRJ_NO
JOIN TASK T
  ON P.DNO = T.DNO
 AND P.PRJ_NO = T.PRJ_ID;
```

16. List the total project budget sanctioned per department.

```
SELECT D.DNAME,
       SUM(P.PRJ_CREDIT) AS total_project_budget
FROM DEPT D
JOIN PROJ P ON D.DNO = P.DNO
GROUP BY D.DNAME;
```

17. List the count of tasks based on their status.

```
SELECT STATUS,
       COUNT(*) AS task_count
FROM TASK
GROUP BY STATUS;
```

18. List employee names in uppercase with joining year.

```
SELECT UPPER(ENAME) AS employee_name,
       TO_CHAR(DATE_JOIN, 'YYYY') AS joining_year
FROM EMP;
```

19. List the number of months each employee has served.

```
SELECT EMPNO,
       ENAME,
       TRUNC(MONTHS_BETWEEN(SYSDATE, DATE_JOIN)) AS months_served
FROM EMP;
```

20. Display the highest-paid employee along with department name for every department.

```
SELECT DNAME, ENAME, SAL
FROM (
    SELECT D.DNAME,
           E.ENAME,
           E.SAL,
           DENSE_RANK() OVER (
               PARTITION BY E.DEPTNO
               ORDER BY E.SAL DESC
           ) AS RNK
    FROM DEPT D
    JOIN EMP E ON D.DNO = E.DEPTNO
)
WHERE RNK = 1;
```

21. List job category and total salary paid for each job category by the company.

```
SELECT JOB AS job_category,
       SUM(SAL) AS total_salary
FROM EMP
GROUP BY JOB;
```

22. Display the department from which the maximum number of employees are working on project P1 (using subquery).

```
SELECT DNAME
FROM DEPT
WHERE DNO = (
    SELECT DEPTNO
    FROM (
        SELECT DEPTNO, COUNT(*) AS employee_count
        FROM EMP
        WHERE PRJ_ID = 'P1'
        GROUP BY DEPTNO
        ORDER BY employee_count DESC
    )
    WHERE ROWNUM = 1
);
```

23. Select name and job of employees working either in Marketing or Research department.

```
SELECT E.ENAME, E.JOB
FROM EMP E
JOIN DEPT D ON E.DEPTNO = D.DNO
WHERE UPPER(D.DNAME) IN ('MARKETING', 'RESEARCH');
```

24. Display name and salary of employees whose salary is greater than minimum salary of the company (using subquery).

```
SELECT ENAME, SAL
FROM EMP
WHERE SAL > (
    SELECT MIN(SAL)
    FROM EMP
);
```

25. Display employees working on the most projects.

```
SELECT EMPNO, ENAME, COUNT(DISTINCT PRJ_ID) AS project_count
FROM EMP
GROUP BY EMPNO, ENAME
HAVING COUNT(DISTINCT PRJ_ID) = (
    SELECT MAX(project_count)
    FROM (
        SELECT EMPNO, COUNT(DISTINCT PRJ_ID) AS project_count
        FROM EMP
        GROUP BY EMPNO
    )
);
```

26. Display the Deptno of the department that has the highest average salary (subquery).

```
SELECT DEPTNO
FROM EMP
GROUP BY DEPTNO
HAVING AVG(SAL) = (
    SELECT MAX(avg_salary)
    FROM (
        SELECT AVG(SAL) AS avg_salary
        FROM EMP
        GROUP BY DEPTNO
    )
);
```

27. List the names of departments which are working on more than 1 project (subquery).

```
SELECT DNAME
FROM DEPT
WHERE DNO IN (
    SELECT DNO
    FROM PROJ
    GROUP BY DNO
    HAVING COUNT(*) > 1
);
```

28. Display how many employees joined after the 15th of all months.

```
SELECT COUNT(*) AS employees_joined_after_15th
FROM EMP
WHERE TO_NUMBER(TO_CHAR(DATE_JOIN, 'DD')) > 15;
```

29. List employee number, name and job of employees who work in the same job as 'Mahesh'.

```
SELECT EMPNO, ENAME, JOB
FROM EMP
WHERE JOB = (
    SELECT JOB
    FROM EMP
    WHERE UPPER(ENAME) = 'MAHESH'
);
```

30. Create a view EMP_PRJ_VW for employees of Marketing department and the project in which they are working.

```
CREATE OR REPLACE VIEW EMP_PRJ_VW AS
SELECT E.EMPNO,
       E.ENAME,
       E.JOB,
       D.DNAME,
       P.PRJ_NO,
       P.PRJ_NAME
FROM EMP E
JOIN DEPT D ON E.DEPTNO = D.DNO
JOIN PROJ P
  ON E.DEPTNO = P.DNO
 AND E.PRJ_ID = P.PRJ_NO
WHERE UPPER(D.DNAME) = 'MARKETING';
```

31. Display employee names and projects using view EMP_PRJ_VW.

```
SELECT ENAME, PRJ_NAME
FROM EMP_PRJ_VW;
```

32. Calculate total and average task duration for each project.

```
SELECT PRJ_ID,
       SUM(E_DATE - S_DATE) AS total_task_duration_days,
       AVG(E_DATE - S_DATE) AS avg_task_duration_days
FROM TASK
GROUP BY PRJ_ID;
```

33. List the number of employees joined in every month in ascending order.

```
SELECT TO_CHAR(DATE_JOIN, 'MM') AS join_month,
       COUNT(*) AS employee_count
FROM EMP
GROUP BY TO_CHAR(DATE_JOIN, 'MM')
ORDER BY join_month ASC;
```

34. Create an index on the columns name and job on EMP table.

```
CREATE INDEX IDX_EMP_NAME_JOB
ON EMP (ENAME, JOB);
```

35. Display the employees working in the same department as Ravi.

```
SELECT *
FROM EMP
WHERE DEPTNO = (
    SELECT DEPTNO
    FROM EMP
    WHERE UPPER(ENAME) = 'RAVI'
);
```

36. Display departments that have at least one project with budget above 400000 (correlated).

```
SELECT D.DNAME
FROM DEPT D
WHERE EXISTS (
    SELECT 1
    FROM PROJ P
    WHERE P.DNO = D.DNO
      AND P.PRJ_CREDIT > 400000
);
```

37. Display employees who earn more than the average salary of their own department (correlated).

```
SELECT E.EMPNO, E.ENAME, E.SAL, E.DEPTNO
FROM EMP E
WHERE E.SAL > (
    SELECT AVG(E2.SAL)
    FROM EMP E2
    WHERE E2.DEPTNO = E.DEPTNO
);
```

38. Display the names of employees who have no dependents (NOT EXISTS).

```
SELECT E.ENAME
FROM EMP E
WHERE NOT EXISTS (
    SELECT 1
    FROM DEPENDENT D
    WHERE D.EMPNO = E.EMPNO
);
```

39. List the names of developers or engineers (SET OPERATION).

```
SELECT ENAME
FROM EMP
WHERE UPPER(JOB) = 'DEVELOPER'
UNION
SELECT ENAME
FROM EMP
WHERE UPPER(JOB) = 'ENGINEER';
```